## === UNIT 2: SENTENCE STRUCTURE, SYNTAX, CONCORD & QUESTION TAGS ===

### >> 2.1 Clause Taxonomy and Four Sentence Architectural Types

Sentence structure dictates the clarity and logical progression of technical arguments.
* Clauses:
- Independent (Main) Clause: Contains a subject and predicate, expressing a complete thought that can stand alone as a grammatical sentence.
- Dependent (Subordinate) Clause: Contains a subject and predicate but begins with a subordinating conjunction (because, although, if, since, when) or relative pronoun, incapable of standing alone.
* The Four Sentence Types:
1. Simple Sentence: Exactly one independent clause: "The enzyme catalyzed the substrate at 37 deg C."
2. Compound Sentence: Two or more independent clauses joined by a coordinating conjunction (FANBOYS) preceded by a comma, or linked by a semicolon: "The reaction reached equilibrium, but the yield remained suboptimal."
3. Complex Sentence: One independent clause combined with at least one dependent clause: "Although the temperature increased, the reaction rate plateaued because the enzyme denatured."
4. Compound-Complex Sentence: At least two independent clauses and one or more dependent clauses: "When the sterile medium was inoculated, cell growth accelerated, and oxygen consumption doubled."
* Common Syntactic Errors:
- Sentence Fragment: Incomplete thought lacking an independent clause.
- Run-On (Fused) Sentence: Two independent clauses joined without punctuation or coordinating conjunctions.
- Comma Splice: Two independent clauses incorrectly joined by only a comma without a conjunction. Fix: replace comma with a semicolon or add coordinating conjunction.

### >> 2.2 Rules of Subject-Verb Agreement (Concord)

A finite verb must agree with its grammatical subject in person and number.
* Rule 1 (Intervening Prepositional Phrases): The verb agrees strictly with the head noun, NOT with intervening nouns inside prepositional phrases:
"The efficacy of the newly developed synthetic antimicrobial peptides IS remarkable" (Efficacy is singular).
* Rule 2 (Compound Subjects): Two singular subjects joined by ’and’ take a plural verb: "DNA and RNA are nucleic acids." Exception: When two singular nouns represent a single combined entity: "Bread and butter is his breakfast."
* Rule 3 (Correlative Proximity Rule): When subjects are connected by ’either...or’ or ’neither...nor’, the verb agrees with the CLOSEST subject:

"Neither the lead researcher nor the laboratory assistants WERE aware of the contamination."
"Neither the laboratory assistants nor the lead researcher WAS aware of the contamination."
* Rule 4 (Indefinite Pronouns): Each, every, either, neither, someone, everybody, anyone are strictly SINGULAR:
"Everybody in the department HAS submitted the annual report."
* Rule 5 (Collective Nouns): Singular when acting as a unified collective unit; plural when individual members act separately:
"The committee HAS reached a consensus" (Unified).
"The committee WERE divided in their opinions" (Individual discord).

### >> 2.3 Question Tags in Scientific Dialogue

A question tag is a mini-question appended to the end of a declarative statement to seek confirmation.
* Fundamental Rules:
1. Polarity Reversal: Positive statement takes a Negative tag; Negative statement takes a Positive tag.
2. Auxiliary Matching: The tag uses the identical auxiliary/modal verb from the main statement. If no auxiliary exists, use appropriate form of 'do' (do, does, did).
3. Pronoun Rule: Tag subject must always be a personal pronoun matching the statement subject.
4. Special Forms: "I am right, AREN'T I?" (Never *amn't I). "Let's begin, SHALL WE?". Imperatives take "WILL YOU?" or "WON'T YOU?".

[EXERCISE 1: Structural Classification of Sentences]
Problem: Classify each of the following sentences into Simple, Compound, Complex, or Compound-Complex, identifying all clauses.
Sentence 1: "The bioreactor maintains sterile conditions through automated filtration."
- Classification: SIMPLE SENTENCE. (Single independent clause; subject: The bioreactor; verb: maintains).
Sentence 2: "The bacterial culture grew rapidly during the logarithmic phase, and the glucose concentration decreased steadily."
- Classification: COMPOUND SENTENCE. (Two independent clauses joined by coordinating conjunction ', and').
Sentence 3: "Because the pH dropped below 5.5, the recombinant protein precipitated out of the aqueous buffer."
- Classification: COMPLEX SENTENCE. (Dependent clause 'Because the pH dropped below 5.5' + Independent clause 'the recombinant protein precipitated...').
Sentence 4: "When the autoclave reached 121 deg C, the timer commenced, and the sterilization cycle ran uninterrupted for twenty minutes."
- Classification: COMPOUND-COMPLEX SENTENCE. (Dependent clause 'When the autoclave reached 121 deg C' + Two independent clauses linked by ', and').

[EXERCISE 2: Subject-Verb Agreement (Concord) Mastery Drill]
Problem: Choose the grammatically correct verb in each of the following technical statements:
Sentence 1: "A cluster of malignant tumor cells (has / have) been identified in the biopsy specimen."
- Correct Verb: HAS. (Head noun is singular ’cluster’; ’of malignant tumor cells’ is an intervening prepositional phrase).
Sentence 2: "Neither the high pressure nor the elevated temperatures (was / were) sufficient to degrade the PTFE gasket."
- Correct Verb: WERE. (In ’neither...nor’, verb agrees with closest subject ’elevated temperatures’ [plural]).
Sentence 3: "Ten milliliters of concentrated hydrochloric acid (was / were) added dropwise to the solution."
- Correct Verb: WAS. (Quantities of measurement, volume, time, and money are treated as singular collective units).
Sentence 4: "The criteria for evaluating the biosafety data (is / are) established by regulatory authorities."
- Correct Verb: ARE. (’Criteria’ is plural; singular form is ’criterion’).

[EXERCISE 3: Comma Splice and Run-On Sentence Correction]
Problem: Identify and repair the comma splices in the following academic excerpts using three distinct syntactic methods:
Faulty Passage: "The primary antibodies were incubated overnight, the secondary fluorescent markers were applied the following morning."
- Repair Method 1 (Semicolon): "The primary antibodies were incubated overnight; the secondary fluorescent markers were applied the following morning."
- Repair Method 2 (Coordinating Conjunction): "The primary antibodies were incubated overnight, and the secondary fluorescent markers were applied the following morning."
- Repair Method 3 (Subordinating Conjunction): "After the primary antibodies were incubated overnight, the secondary fluorescent markers were applied the following morning."

[EXERCISE 4: Advanced Question Tag Construction]
Problem: Supply the correct question tag for each of the following technical communication utterances:
Sentence 1: "The technician calibrated the analytical balance before the measurement, _____?"
- Solution: DIDN’T HE? / DIDN’T SHE? / DIDN’T THEY? (Simple past ’calibrated’ requires auxiliary ’did’; positive statement takes negative tag ’didn’t’).
Sentence 2: "None of the recombinant clones exhibited resistance to ampicillin, _____?"
- Solution: DID THEY? (’None’ is negative, requiring a positive tag; pronoun for clones is ’they’).
Sentence 3: "I am responsible for maintaining the cleanroom environmental logs, _____?"

- Solution: AREN'T I? (Standard idiomatic question tag for 'I am').
Sentence 4: "Let us review the chromatography data once more, _____?"
- Solution: SHALL WE? ('Let's / Let us' suggestions mandate 'shall we').

### >> 2.4 University Review & Examination Questions

* Q1: State five major rules of Subject-Verb Agreement (Concord) with original illustrative sentences from biotechnology.
Answer: (1) Intervening phrases: Verb agrees with head noun, ignoring intervening prepositional modifiers: 'The stability of multiple cloned plasmids IS monitored weekly.' (2) Compound subjects joined by 'and' take plural verbs: 'Centrifugation and electrophoresis ARE analytical separation techniques.' (3) Either...or / Neither...nor proximity rule: Verb agrees with the nearest noun: 'Neither the primer nor the nucleotide reagents WERE contaminated.' (4) Indefinite pronouns (each, every, either, neither) are singular: 'Each of the PCR cycles REQUIRES precise thermal regulation.' (5) Measurements of quantity, volume, and time take singular verbs: 'Fifty milligrams of lyophilized enzyme WAS dissolved in buffer.'
* Q2: Explain the Comma Splice error and discuss three distinct syntactical methods to resolve it.
Answer: A comma splice occurs when two independent clauses (complete grammatical sentences) are joined together solely by a comma without an accompanying coordinating conjunction: 'The cells reached confluence, the media was changed.' This is ungrammatical because a comma has insufficient structural weight to link independent sentences. Three resolution methods: (1) Semicolon: Replace the comma with a semicolon when the two thoughts are closely linked logically: 'The cells reached confluence; the media was changed.' (2) Coordinating Conjunction (FANBOYS): Insert a coordinating conjunction after the comma: 'The cells reached confluence, so the media was changed.' (3) Subordinating Conjunction: Convert one independent clause into a dependent clause: 'When the cells reached confluence, the media was changed.'

### >> 2.5 Key Takeaways and Revision Points

* Compound sentences join independent clauses with coordinating conjunctions; complex sentences combine independent and dependent clauses.
* Comma splices occur when two independent clauses are linked by a comma alone; fix with semicolon or conjunction.
* Concord requires subject-verb agreement; intervening prepositional phrases do not alter subject number.
* Question tags reverse statement polarity and utilize matching auxiliary verbs (e.g., 'I am late, aren't I?').